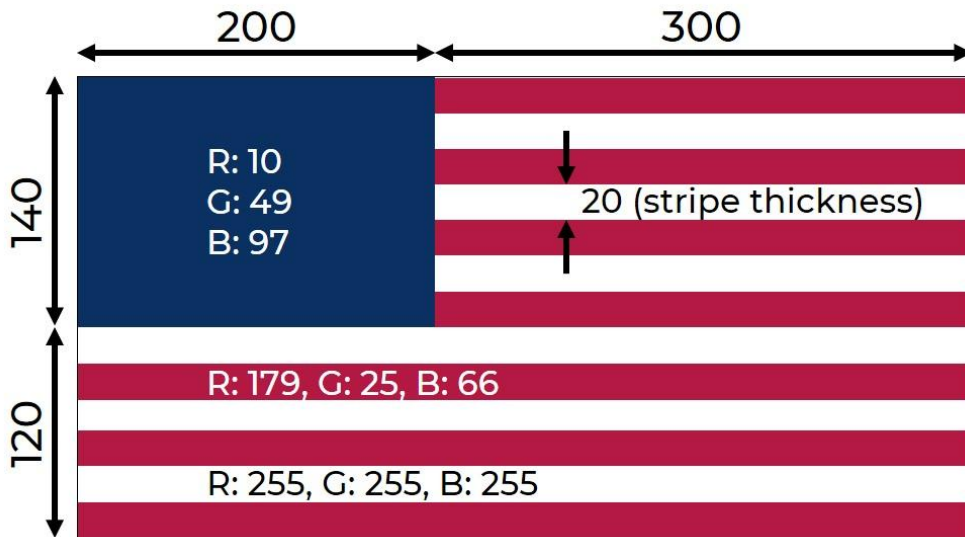


Lesson 7: Creating Complex Color Filters

Creating Complex Color Filters

Let's test out new concatenation skills by creating a complex filter matching the specifications of the national flag of the United States of America (except without the 50 stars). We'll once again use the `newFilter` function from the [Pixels to Pictures](#) toolbox to create each color stripe, and then use concatenation to stitch the pieces together.



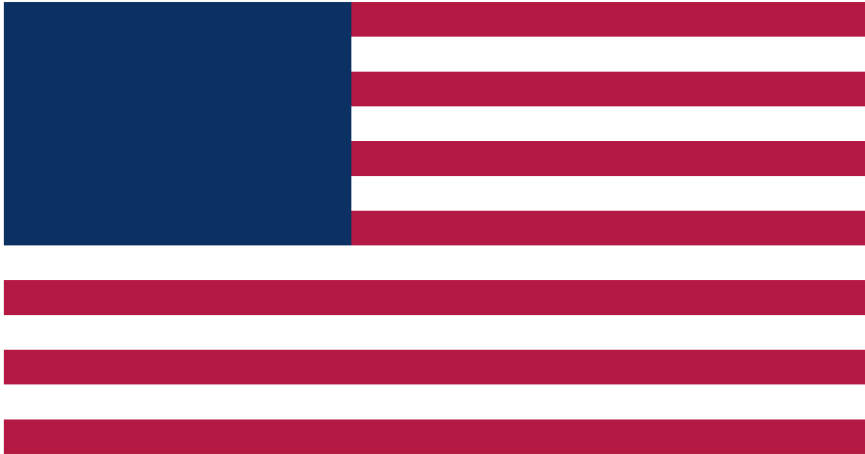
```
% Use the newFilter function to create the blue field
BF = newFilter(140,200,[10 49 97]);

% Use the newFilter function to create red and white short stripes
RS = newFilter(20,300,[179 25 66]);
WS = newFilter(20,300,[255 255 255]);
SS = [RS; WS; RS; WS; RS; WS; RS]; % vertically concatenate 7 short stripes

% Use the newFilter function to create red and white long stripes
RL = newFilter(20,500,[179 25 66]);
WL = newFilter(20,500,[255 255 255]);
LS = [WL; RL; WL; RL; WL; RL]; % vertically concatenate 6 short stripes

% Use square brackets and semicolons to concatenate filters pieces together
Flag1 = [BF SS; LS];

% Show concatenated filter
imshow(Flag1)
```



Add Filter to an Image

The url below links to an encyclopedia image of the Statue of Liberty in New York Bay. Or, feel free to use images from the [Pixels to Pictures Image Library](#) or your own images. If you use your own images, just make sure to provide the full path to the image so MATLAB can find it.

```
url = "https://cdn.britannica.com/61/93061-050-99147DCE/" ...  
    + "Statue-of-Liberty-Island-New-York-Bay.jpg";  
Img = imread(url);           % read an image  
sz = size(Flag1);            % get size of filter  
Img = imresize(Img,sz(1:2)); % resize image to match  
  
Combo1 = Img + 0.5*Flag1;    % reduce filter intensity by half  
imshow(Combo1)              % display combined image/filter
```

Use a Flag Image Instead

Recall our color filters are just the concatenation of single color images. So, now that we've learned about concatenation, we could just use an image of a flag directly instead of creating a complex filter.

```
url = "https://cdn.britannica.com/33/4833-050-F6E415FE/" ...  
    + "Flag-United-States-of-America.jpg";  
Flag2 = imread(url);  
Flag2 = imresize(Flag2,sz(1:2)); % resize to match image  
imshow(Flag2)                   % display flag image  
  
Combo2 = Img + 0.5*Flag2;      % reduce filter intensity by half  
imshow(Combo2)                % display combined image/filter
```